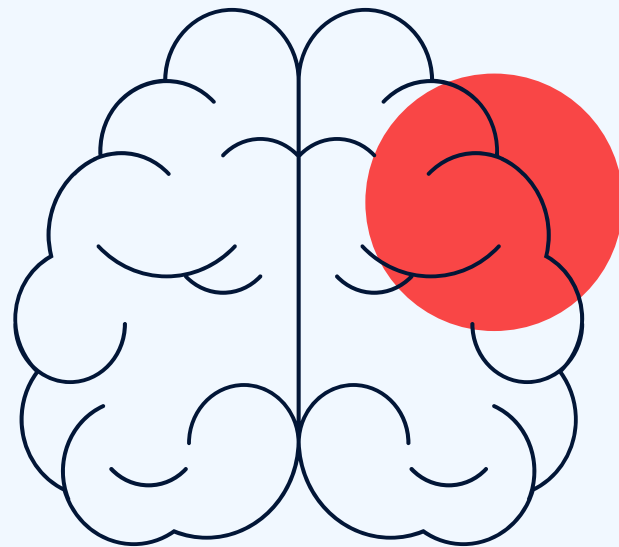


Stroke Prediction

By: Daphne O'Malley, Maxamed Jama, Zoe Cheng, Ashay Srivastava,
Allen Liu, and Shemar Anglin



Research Question

Can we predict whether a patient will have a stroke (Yes or No)
based on their clinical and lifestyle information?

Yes (1) → The patient is likely to have a stroke

No (0) → The patient is not likely to have a stroke

Background/Importance of the Research

- **Why is this health issue important?**

- Stroke is the second leading cause of death in the world.
- It is the third leading cause of death and disability combined (calculated by disability-adjusted life-years lost).

Source: [World Stroke Organization \(2022\)](#)

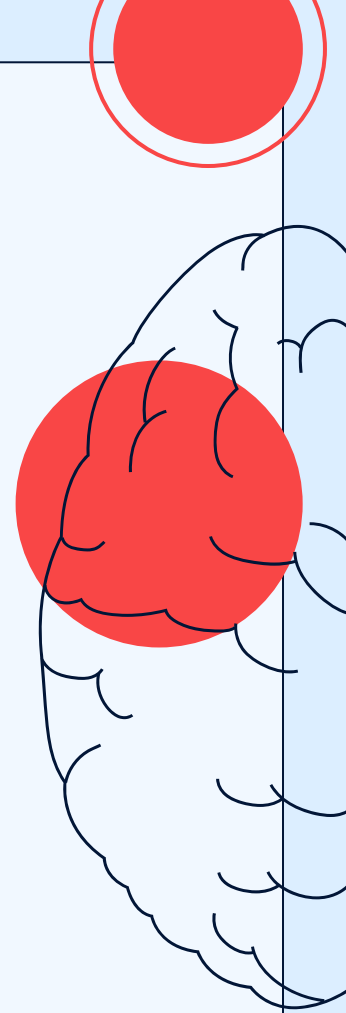
- **Who might benefit from this kind of prediction model.**

- Healthcare Providers: With early detection methods, healthcare professionals can intervene quickly for patients who are high-risk.
- Patients: Assessments of patients and their risk behaviors can help to make health and lifestyle decisions to reduce the risk of stroke.

- **Statistics of real-world use cases.**

- In the United States:
 - Every 40 seconds someone experiences a stroke.
 - Every 3 minutes and 11 seconds, someone dies of a stroke.
 - Annually, at least 795,000 people have a stroke.

Source: [Center for Disease Control and Prevention](#)



Data Cleaning & Preprocessing

Source: Stroke Prediction Dataset ([strokepredictiondataset](#)) - Kaggle
There are 5110 observations & 12 variables

Data preprocessing steps taken:

1. Import & Format

- Loaded the stroke data set, using `read.csv()`
- Converted the “stroke” variable into a factor, labeled: “yes” and “no”

2. Addressed Missing Data

- Replaced “N/A” with NA; as recognized in R
- Replace missing values in “bmi” column with the median

3. Summary

- Ran “`summary()`” to analyze the variable distributions

4. Encode Categorical Data

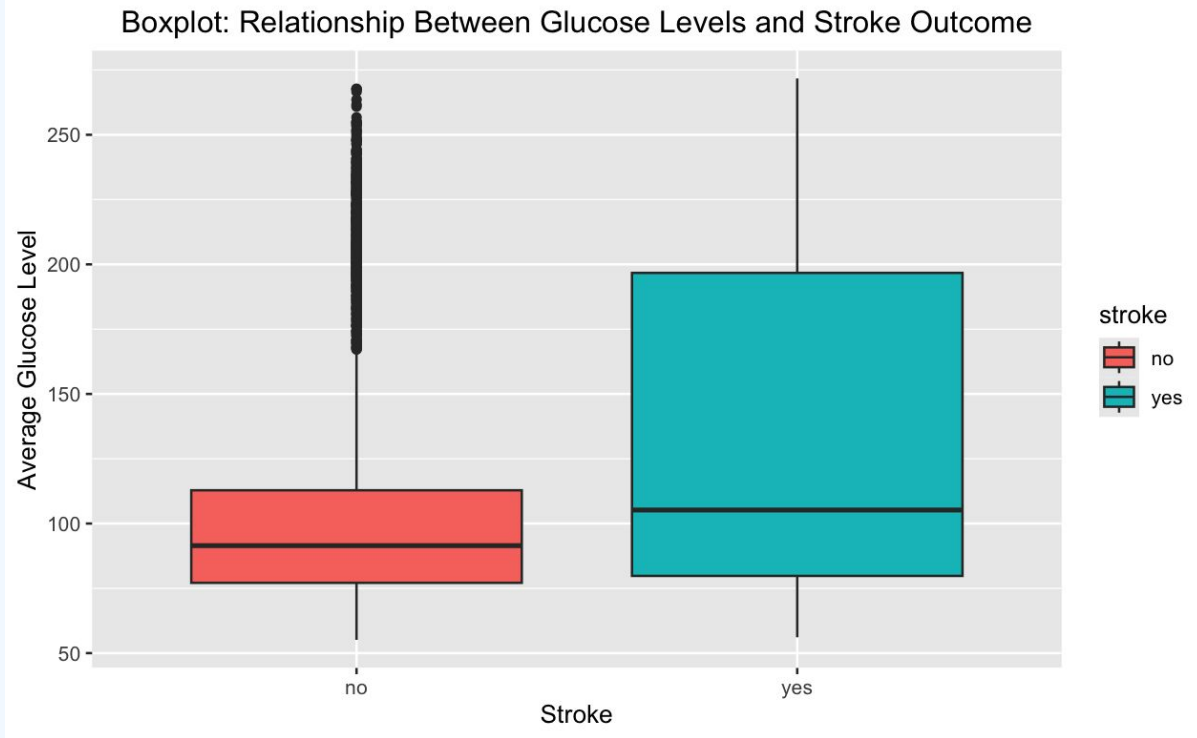
- One-hot encoded: gender, ever_married, work_type, residence_type, smoking_status
- Used `model.matrix()` to perform a loop for converting each category to numeric
- Dropped original columns

5. Final Check

- Verify that all columns are ready for modeling
- Confirmed no missing values in the encoded dataframe

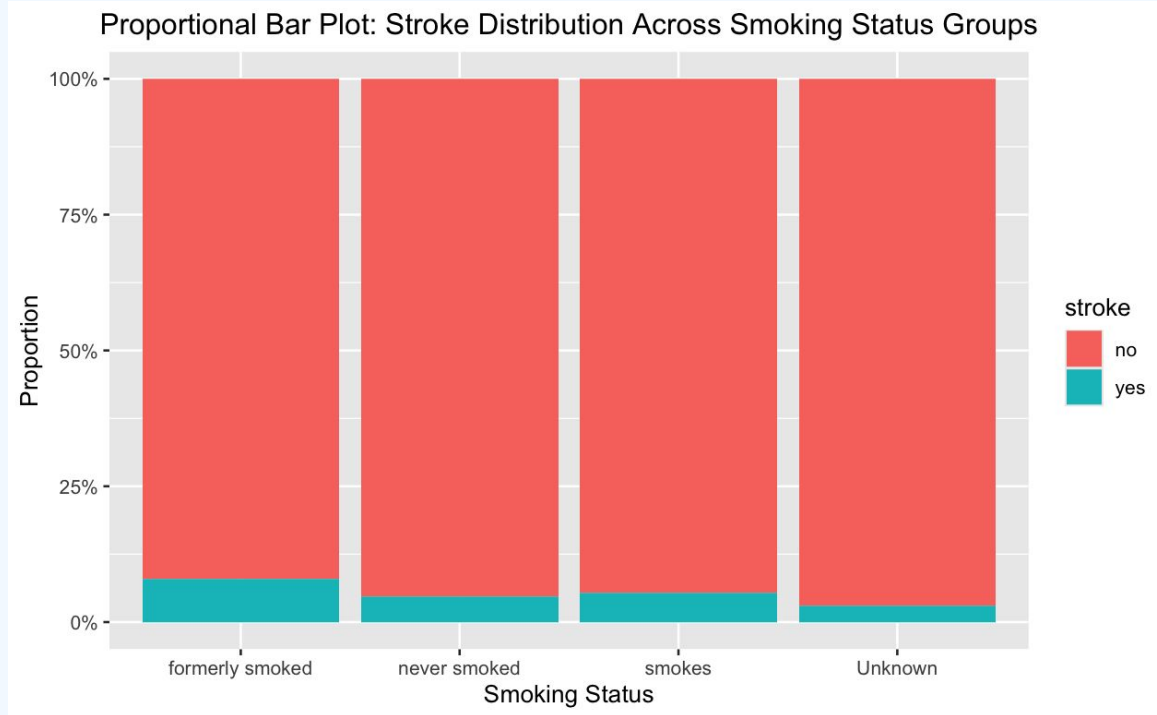


Data Visualization 1 - Glucose vs. Stroke



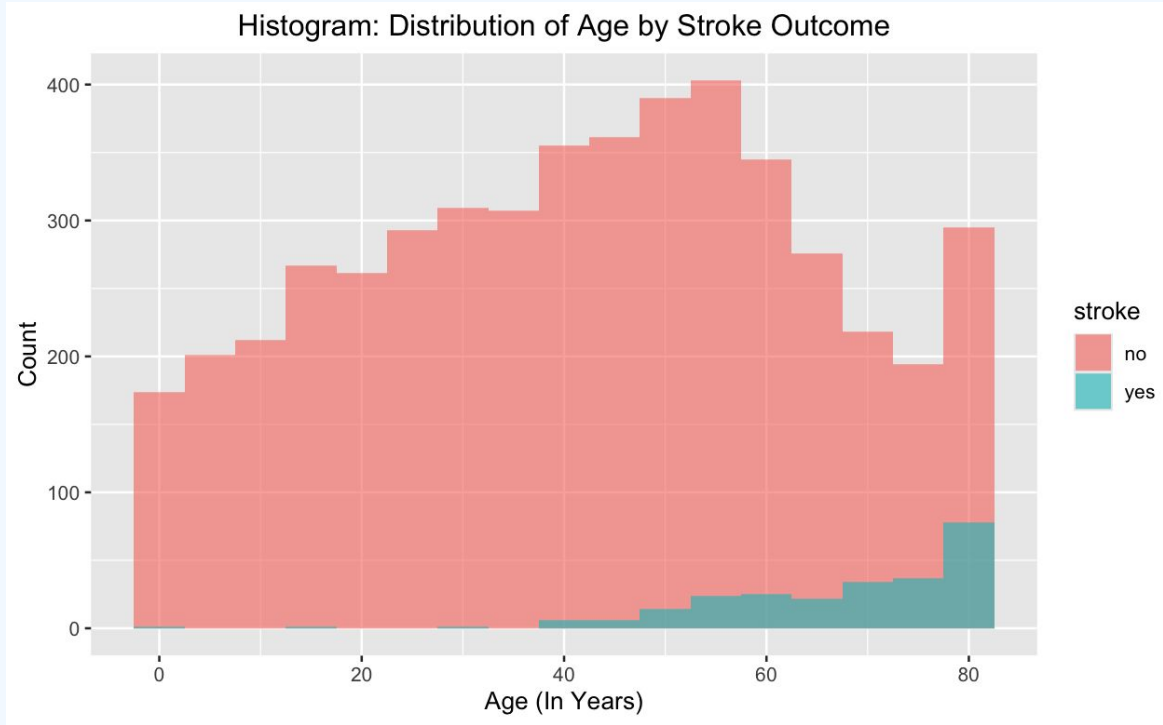
- Stroke patients have a wider and higher range of glucose levels.
- The outliers for non-stroke patients suggest that some patients still had very high glucose levels.
- This plot suggests that glucose levels are a strong clinical predictor for stroke modeling.

Data Visualization 2 - Stroke by Smoking



- The stroke rate is highest for those who currently smoke and for those who used to smoke.
- Patients in “never smoked” have the lowest proportion of strokes (excluding ‘unknown’ group).
- This plot shows the relevance of smoking status/behavior as a risk factor.

Data Visualization 3 - Age by Stroke



- There is a significant increase in stroke cases in patients aged 50+.
- Younger age groups show little to no stroke occurrences.
- This plot shows that age is a major contributor to predicting stroke events.

Cross-Validation Model Performance

Models: DecisionTree, SVM, kNN

Number of resamples: 5

ROC

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
DecisionTree	0.5000000	0.6803841	0.7091925	0.6886105	0.7374377	0.8160383	0
SVM	0.5691647	0.6641086	0.6641827	0.6491640	0.6643836	0.6839803	0
kNN	0.5939087	0.6206381	0.6322468	0.6392175	0.6540985	0.6951952	0

Sens

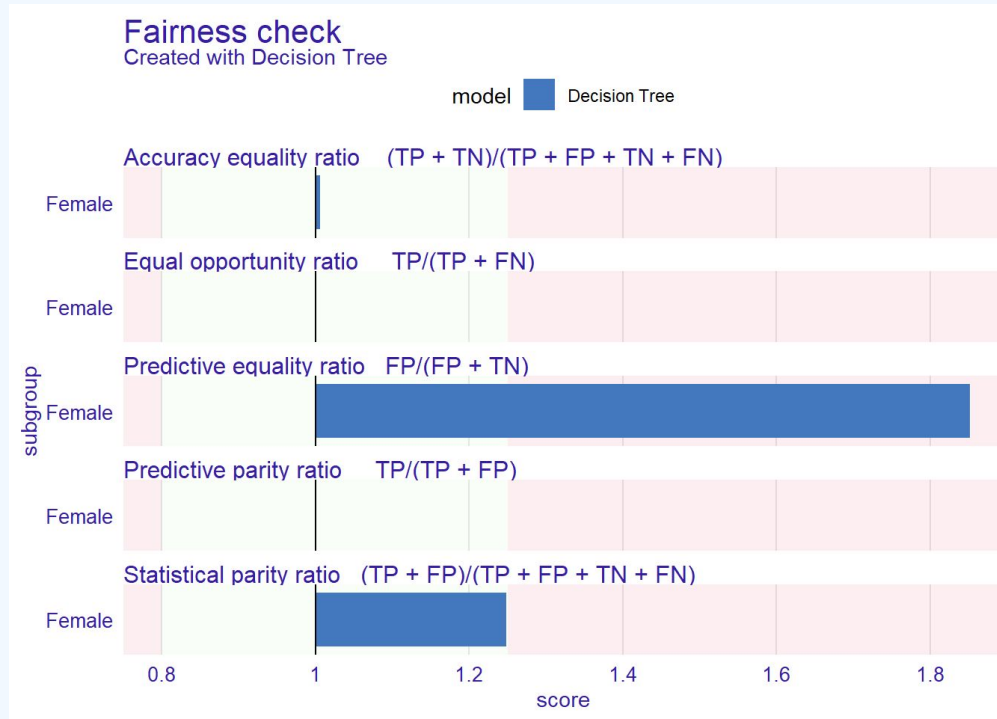
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
DecisionTree	0.9821674	0.9863014	0.9903978	0.9909503	0.9958848	1.0000000	0
SVM	0.9972603	0.9986283	1.0000000	0.9991777	1.0000000	1.0000000	0
kNN	0.9931413	0.9945130	0.9945130	0.9950629	0.9958904	0.9972565	0

Spec

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
DecisionTree	0	0.02631579	0.02702703	0.03726885	0.05405405	0.07894737	0
SVM	0	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000	0
kNN	0	0.00000000	0.02631579	0.02147937	0.02702703	0.05405405	0

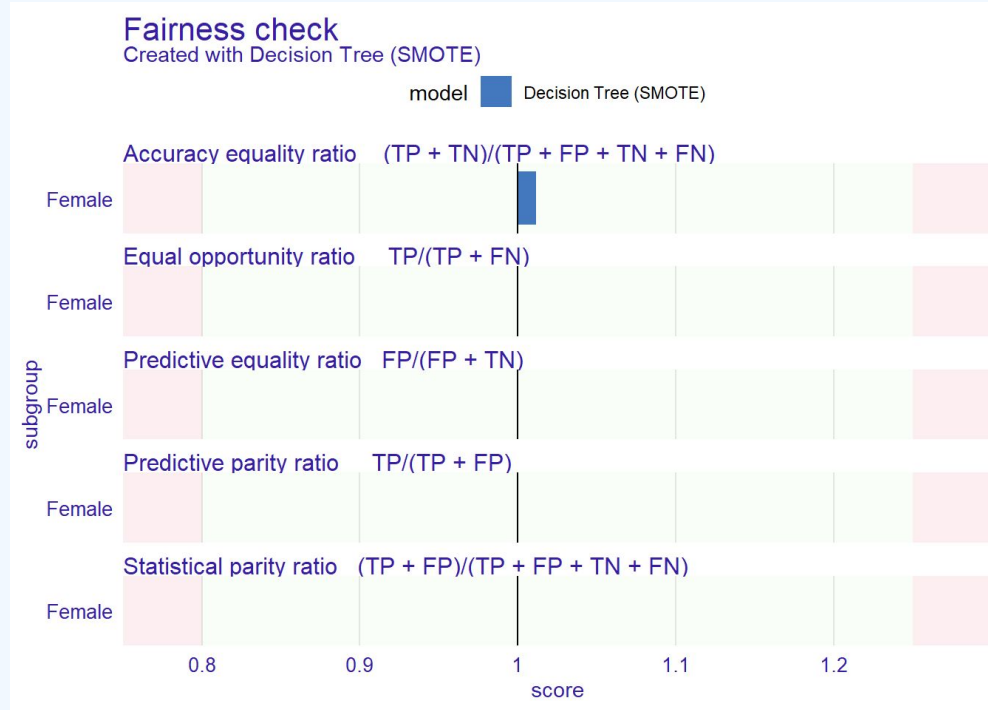
- The SVM and k-NN models have very high sensitivity and very low specificity indicating a high case of false alarms.
- The Decision Tree model has the highest ROC score (0.69) as well as a good balance between sensitivity and specificity.
- The Decision Tree is the best model.

Fairness and Bias Evaluation



- Bias was detected for the female group. With high disparity for the Predictive Equality Ratio and the Statistical Parity Ratio
- This means that the model had a high chance of giving false diagnosis among females, which is very risky from a clinical standpoint.

Fairness and Bias Evaluation



- SMOTE was applied to balance the stroke classes within the training data.
- After SMOTE was used, the ratios of fairness came closer to 2, which showed a significant reduce in bias for gender.
- SMOTE was the best option because of its low sensitivity AND class imbalance, as described in the course.

Test Set Evaluation



- Each model shows high accuracy, which means that predicted stroke outcomes are overall reliable.
- Decision Tree had the highest ROC, meaning that this model was best at making distinctions between true positive and false positive rates for distinguishing stroke vs. no stroke patients.
- Sensitivity was extremely low, which meant that the models had trouble identifying actual stroke cases for patients.

Metric Selection

Chosen Metric: Sensitivity

- Sensitivity measures the models' ability to identify true stroke cases for patients.

Why Sensitivity Matters?

- Clinically speaking, if healthcare providers were to miss true stroke cases (false negatives), this can lead to a lack of proper intervention for patients who are at risk. This can lead to severe disability or even death.

Real-World Implications

- Medical professionals are concerned more with mitigating stroke, rather than dealing with it after the event occurs.
- Sensitivity ensures that they are able to catch as many high-risk patients that they can, regardless of them being false alarms.
- For stroke care, it's better for patients and medical professionals to over-identify the risk of a stroke than missing a potential stroke.



Recommendation to Clinicians

Recommendation: Due to the low sensitivity of each model, none of them should be used clinically.

Reasoning: Models that failed to identify stroke cases correctly would be dangerous to use in real-world clinical environments. This risks not alerting the appropriate people when urgent care regarding stroke is needed.

Possible Improvements:

- Improve models for better detection of stroke cases.
 - Try alternative models to improve performance beyond kNN and/or SVM models.
 - Trying different inputs for parameters like k for kNN or cost in SVM.
 - Reassess any data preprocessing steps.
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